

probability distribution using Platt scaling through cross-validation on the training data (Platt, 1999).

Statistics-based Probes. For statistics-based probes, we present logistic regression (LR) and multi-layer perceptron (MLP). LR is commonly used as a baseline, while MLP is termed SAPLMA by Azaria and Mitchell (2023).

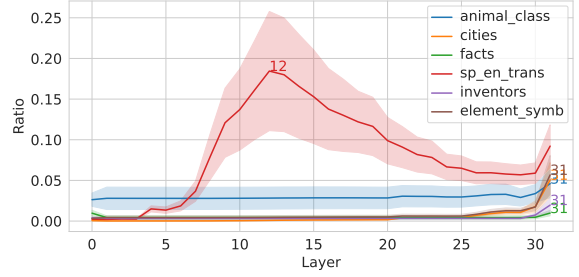
3.3 Data for Probing Truthfulness

The binary probes introduced above are applied to the truthfulness classification task, assuming the availability of truthfulness-specific data. We use the factual statements curated by Bürger et al. (2024), drawing from datasets by Azaria and Mitchell (2023) and Marks and Tegmark (2023). These datasets cover a variety of topics, including `animal_class`, `cities`, `element_symb`, `facts`, `inventors`, `sp_en_trans`, as well as variations incorporating logical negations, conjunctions and disjunctions. Each statement is labeled as “true” or “false”, indicating its factuality. Statements can be atomic or compound. Atomic statements make individual claims, either affirmative or negative. Negative statements correspond to their affirmative counterparts, with syntax-level negation and inverted labels. Compound statements are created by logically combining atomic statements of the same topic through conjunction or disjunction.

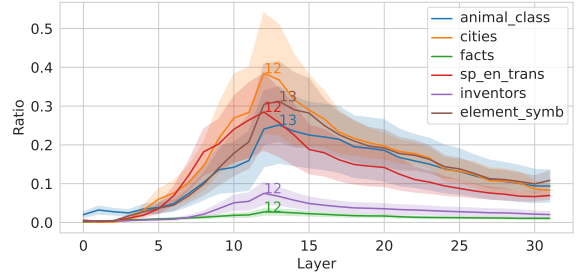
4 Experiments

4.1 Preliminary Experiment: Layer Selection

Identifying the optimal layer for detecting truthfulness is crucial for probe performance. Marks and Tegmark (2023) observed that the truth direction “emerges rapidly in early-middle layers”; however, this observation does not indicate which specific layer provides the most effective representations. To address this, we adopt the technique used by Bürger et al. (2024) and MacDiarmid et al. (2024), which evaluates the difficulty of separating true/false statements across layers by analyzing variance. The ideal layer maximizes the separation between true and false representations, quantified by the “between-class variance”, relative to the internal variance within each class, referred to as “within-class variance”. By plotting the ratio of between-class to within-class variance across decoder layers for a range of topic-specific datasets, we identify the optimal layer as the one with the highest ratio.



(a) Llama-2-7B



(b) Llama-3.1-8B

Figure 2: Ratio of between-class variance to within-class variance across layers. The layer indices (starting from 0) for the greatest ratios are annotated at the summit of each curve. The solid curves are mean values, and the surrounding shades denote standard error.

We present the ratio of between-class variance to within-class variance across layers for Llama-2-7B (Touvron et al., 2023) and Llama-3.1-8B (Dubey et al., 2024) in Figure 2, with results for additional models in the Appendix. Each curve represents statements involving affirmations, negations, conjunctions and disjunctions of the same topic. For Llama-3.1-8B, the 12th layer (zero-indexed) emerges as the optimal layer. In contrast, for Llama-2-7B, a peak occurs only for the `sp_en_trans` topic, with minimal separation observed for other topics. This suggests that, while Llama-2-7B may internally represent truthfulness as a feature, it does so in a domain-specific manner, with limited consistency across knowledge domains. Additionally, this feature appears to lack salience in the early-middle layers.

4.2 Probing a Randomized Model

This section investigates whether the truth direction is an inherent structure within a pretrained LLM or an artifact constructed by the truthfulness probe. To address this, we randomly initialize the weights of Llama-3.1-8B, extract activations from its 12th layer using the `animal_class` dataset, and train probes on a 70% split while testing them on the

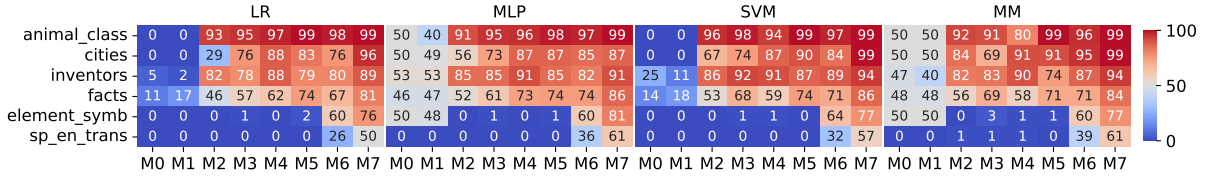


Figure 3: AUROC (in percentage) of probes trained on affirmative statements and tested on negative ones. AUROC > 0.5 indicates success of generalization. M0-M7 refer to models from Llama-2-7B to Llama-3.1-70B-Instruct.

remaining 30%. Results show that the AUROCs are 0.50, 0.52, 0.58, 0.50 for the LR, MLP, MM and SVM probes, respectively. In contrast, when using the pretrained weights, the AUROCs achieve 1.0 across all probes. These findings demonstrate that the simple probes introduced in Section 3.2 cannot independently construct a truth direction, confirming that truth direction is a product of the pretraining process.

4.3 Consistency of Truth Directions

Levinstein and Herrmann (2024) claim that probes such as MLP fail to generalize across negation. However, we question this conclusion and hypothesize that the generalization performance of truthfulness probes is influenced more by the targeted LLM than by the probe itself. To test this hypothesis, we examine whether the truth direction identified for affirmative statements is consistent with that identified for negative statements of the same topic.

4.3.1 Experimental Setup

Data. For each of the six topics introduced in Section 3.3, we train probes on affirmative statements and test them on corresponding negative ones. For example, we train probes on affirmative statements of the `animal_class` topic, and test them on `neg_animal_class`. Note that the training and test data contain the same set of knowledge, differing only in syntax.

Models. We select a series of LLMs with increasing levels of general capability, including both foundational models and instruction-tuned ones (According to the evaluation results on a range of standard benchmarks²): Llama-2-7B(-Chat), Llama-2-13B(-Chat), Llama-3.1-8B(-Instruct), Llama-3.1-70B(-Instruct). Their optimal layers are 12(13), 13(13), 12(13), 33(33), respectively.

Methods. We employ the four probe instantiations introduced in Section 3.2.

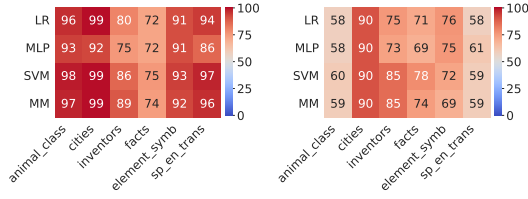
²<https://github.com/meta-llama/llama-models/blob/main/models>

Metrics. The metric used is AUROC (Area Under Receiver Operating Characteristic Curve) which is commonly used to assess classifier performance. A probe is considered to generalize successfully on a topic if its test AUROC exceeds 0.5, indicating performance better than chance. Results are averaged across three trials, with randomness introduced through probe initialization and data splits for cross-validation of Platt scaling, while the training data remains constant.

4.3.2 Results

The results are presented in Figure 3. We observe a positive correlation between the performance of truthfulness probes and the general capability of the target models. For Llama-2-7B(-Chat), the probes fail to generalize on all six topics. For Llama-2-13B(-Chat) and Llama-3.1-8B(-Instruct), the probes generalize on four topics; for Llama-3.1-70B they generalize on five topics; and for Llama-3.1-70B-Instruct, they generalize on all six topics.

Regarding whether the truthfulness probe is a faithful reflection of the actual truth direction, we borrow the *Weak-to-Strong Explanation* from Zhou et al. (2024): if weak classifiers can successfully distinguish the representations, it indicates that LLMs have implicitly converted inputs to different representations. For the most capable model, Llama-3.1-70B-Instruct, all the simple probes we use are able to generalize across logical negation on all six topics. This suggests that Llama-3.1-70B-Instruct consistently represents truthfulness in its internals for both affirmative and negative statements. Therefore the results suggest a potential correlation between the degree of generalization of its truth direction and the model’s capability (e.g., knowledge capacity and natural language understanding ability). Additionally, the differences in performance between probes become negligible starting with Llama-2-13B-Chat and onward. This indicates that, for more capable LLMs, probe performance is more influenced by the target model itself than by the design of the probes.



(a) Logical conjunctions. (b) Logical disjunctions.

Figure 4: AUROC (in percentage) of probes trained on atomic factual statements and tested on logical conjunctions/disjunctions for Llama-3.1-8B. AUROC > 0.5 indicates the success of generalization.

4.4 Binary Logical Transformation

In Section 4.3, we tested the ability of truthfulness probes to generalize across logical negation. In this experiment, we extend the analysis to more complex binary logical transformations, specifically logical conjunction and disjunction. This requires the LLM to perform several implicit tasks: identify the truthfulness of both atomic statements, interpret binary logical operators from natural language to abstract concepts, and apply the operators to compute the joint truthfulness.

4.4.1 Experimental Setup

Data. The training data consists of all atomic statements. The test data comprises logical conjunctions and disjunctions of atomic affirmative statements for each topic. The knowledge of the test data is covered by the training data.

Models. We use Llama-3.1-8B as the primary model for demonstration.

Methods. We apply the four probe instantiations introduced in Section 3.2.

Metrics. For this experiment, we use AUROC as a measure of probe performance. A probe is considered to generalize successfully on a topic if its test AUROC exceeds 0.5. Results are averaged across three trials, with randomness introduced through probe initialization and training data splits.

4.4.2 Results

According to Figure 4, all probes successfully generalize to both logical conjunctions and disjunctions. However, performance is notably stronger for logical conjunctions compared to disjunctions across animal_class, element_symb, and sp_en_trans topics. This discrepancy may suggest that disjunctions pose a greater challenge for Llama-3.1-8B to interpret truthfulness.

4.5 Question Answering

We hypothesize that if truthfulness is consistently represented in an LLM’s internal states, this representation should depend solely on the semantics of a sentence, rather than its syntax form. Additionally, question answering is more common than statements in real-world human-AI interactions. Motivated by these considerations, we examine if truthfulness probes, trained on atomic factual statements, can generalize to the QA setting.

We test on a multiple-choice task and a short-form QA task. We also investigate the in-context learning scenario, a popular prompting technique for teaching LLMs new tasks at inference time. While in-context examples can be beneficial, they may include incorrect or misleading examples, which raises questions about how probes handle false examples. Therefore, we pay special attention to the behavior of the probes when incorrect examples are present.

4.5.1 Experimental Setup

Data. The training data consists of all atomic statements. The test data includes MMLU (Hendrycks et al., 2020) and TriviaQA (Joshi et al., 2017). For MMLU, we sample 50 questions from the test set for each of the 57 sub-tasks. As it is a multiple-choice dataset, for each question we select the correct answer and an incorrect answer. For TriviaQA, we sample 20 answers per question from the model at unit temperature.

Models. We use Llama-3.1-8B, as it is a high-capability LLM with ~ 10 B parameters.

Methods. We apply the four probe instantiations introduced in Section 3.2.

Metrics. In addition to classification accuracy, we evaluate calibration, as it is crucial for assessing the reliability of predictions. Specifically, we evaluate AUROC, Expected Calibration Error (ECE) and Brier Score (BS). ECE measures calibration, while BS reflects both accuracy and calibration. Lower values are preferred for both ECE and BS. For ECE we use a binned approach with 10 bins, where each bin contains an equal number of samples, and report the mean absolute error between the accuracy and confidence within each bin. The random baseline for BS is 0.25, corresponding to a uniform prediction of 0.5. Results are averaged across three trials, with randomness introduced via probe initialization and training data splits.